Larkhall NDE Services Inc.

Inspection Report

Site: Hamilton Plant

No.	Status	Date
LNDE-RPT-20241025-031-R1	Issued	2024-11-21

UT Thickness Survey - ATU-100 Five-Year

Scope and Objectives

This report details the five-year ultrasonic thickness survey conducted at the Hamilton Plant for the ATU-100. The survey was initiated following QSCP-RFQ-2024-089 to assess the current wall thickness and remaining life of 18 piping lines. This final revision, LNDE-RPT-20241025-031-R1, supersedes LNDE-RPT-20241025-031-R0 and incorporates corrections requested in EML-CA-20241119-a regarding CML 4-RA-102-05, 3-RF-101-04, and 6-SG-102-04.

The inspection scope covers 150 CMLs across various process services including sour gas inlet lines 6-SG-101 and 6-SG-102, treated gas line 8-TG-101, and lean amine supply lines 4-LA-101 and 4-LA-102. The survey also includes lean amine header 6-LA-103, rich amine lines 4-RA-101 and 4-RA-102, rich amine header 6-RA-103, and regenerator overhead lines 3-RV-101 and 2-RV-102. Additional lines surveyed include reflux lines 3-RF-101 and 2-RF-102, exchanger lines 4-LA-104 and 4-RA-104, reboiler steam line 2-ST-101, and cooling water lines 3-CW-101 and 3-CW-102.

Personnel and Procedures

The field measurements were performed by Marcus Chen, lead UT technician, qualified to ASNT SNT-TC-1A UT Level II. The technical review and quality oversight were provided by David Park, acting as Level III / Quality Manager under ASNT SNT-TC-1A. All activities were executed in accordance with QSCP-NDT-001 Rev 3, which aligns with QSCP-CIS-001 Rev 1 requirements for digital export and CML-id formatting.

Inspection methodologies adhered to API 570 for piping inspection and ASTM E797 for manual ultrasonic thickness measurement. The piping systems were originally designed to ASME B31.3. Data comparison was performed against the previous re-baseline survey recorded in LNDE-RPT-20191001-027-R0 to determine long-term and short-term corrosion rates.

Equipment and Calibration

Measurements were obtained using Olympus 38DL PLUS gauges, identified by LNDE-CAL-2024-0847 and LNDE-CAL-2024-0848. These instruments were calibrated on 2024-10-15 specifically for this five-year survey. Field verification was performed using the carbon steel step block LNDE-CS-001, which was re-certified under LNDE-CAL-2024-0102.

For the stainless steel reflux lines 3-RF-101 and 2-RF-102, the gauges were adjusted to the appropriate stainless-steel velocity setting. In compliance with QSCP-NDT-001 Rev 3, digital photos of the gauge display were captured for all readings that fell below the established alert levels.

Inspection Findings and Analysis

The survey identified five CMLs where the measured wall thickness has reached the retirement limit. On the rich amine line 4-RA-101, CML 06 measured 3.59 mm, which is below the t-min of 3.81 mm. On the reflux line 2-RF-102, four locations (CMLs 04, 05, 06, and 07) recorded thicknesses between 1.86 mm and 1.94 mm, all falling below the t-min of 2.16 mm. These locations are considered expired and require immediate attention per site rules.

Remaining life calculations were performed per QSCP-IP-ATU100 Rev 3, utilizing the larger of the long-term and short-term corrosion rates. While Class 1 lines like 6-SG-101 and 3-RV-101 generally follow a 3-year RBI interval, the identified thinning on the Class 2 reflux and rich amine lines necessitates a revision of the next inspection date to ensure compliance with the half-life rule.

Thickness readings, measured 2024-11-04

Line 6-SG-101 | Sour gas inlet | A106 Gr.B | t_nom 7.11 mm | t-min 3.56 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
6-SG-101-01	7.11	3.56	2022-09-16	5.82	5.64	0.052	0.057	36.2	2027-11-04	
6-SG-101-02	7.11	3.56	2022-09-16	5.42	5.12	0.067	0.077	20.2	2027-11-04	
6-SG-101-03	7.11	3.56	2022-09-16	5.29	5.25	0.064	0.061	26.5	2027-11-04	
6-SG-101-04	7.11	3.56	2022-09-16	5.19	4.98	0.074	0.071	19.3	2027-11-04	
6-SG-101-05	7.11	3.56	2022-09-16	4.72	4.65	0.087	0.077	12.5	2027-11-04	

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
6-SG-101-06	7.11	3.56	2022-09-16	5.55	5.39	0.059	0.034	31.2	2027-11-04	
6-SG-101-07	7.11	3.56	2022-09-16	6.16	6.04	0.040	0.040	>50	2027-11-04	
6-SG-101-08	7.11	3.56	2022-09-16	5.57	5.54	0.058	0.046	34.4	2027-11-04	

Line 6-SG-102 | Sour gas inlet | A106 Gr.B | t_nom 7.11 mm | t-min 3.56 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
6-SG-102-01	7.11	3.56	2022-09-16	4.77	4.62	0.088	0.085	12.1	2027-11-04	
6-SG-102-02	7.11	3.56	2022-09-16	4.64	4.59	0.088	0.101	10.2	2027-11-04	
6-SG-102-03	7.11	3.56	2022-09-16	4.69	4.45	0.090	0.093	9.6	2027-11-04	gauge photo attached
6-SG-102-04	7.11	3.56	2022-09-16	4.62	4.39	0.091	0.101	8.2	2027-11-04	gauge photo attached
6-SG-102-05	7.11	3.56	2022-09-16	5.96	5.89	0.043	0.036	>50	2027-11-04	
6-SG-102-06	7.11	3.56	2022-09-16	5.79	5.82	0.043	0.042	>50	2027-11-04	

Line 8-TG-101 | Treated gas | A106 Gr.B | t_nom 8.18 mm | t-min 4.06 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
8-TG-101-01	8.18	4.06	2019-10-18	7.63	7.57	0.022	0.012	>50	2030-11-04	
8-TG-101-02	8.18	4.06	2019-10-18	7.33	7.21	0.035	0.024	>50	2030-11-04	
8-TG-101-03	8.18	4.06	2019-10-18	7.37	7.27	0.033	0.020	>50	2030-11-04	
8-TG-101-04	8.18	4.06	2019-10-18	7.38	7.29	0.028	0.018	>50	2030-11-04	
8-TG-101-05	8.18	4.06	2019-10-18	7.57	7.54	0.024	0.006	>50	2030-11-04	
8-TG-101-06	8.18	4.06	2019-10-18	7.75	7.61	0.028	0.028	>50	2030-11-04	
8-TG-101-07	8.18	4.06	2019-10-18	7.17	6.98	0.047	0.038	>50	2030-11-04	

Line 4-LA-101 | Lean amine supply | A106 Gr.B | t_nom 6.02 mm | t-min 3.18 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-LA-101-01	6.02	3.18	2019-10-18	4.70	4.46	0.053	0.048	24.0	2030-11-04	
4-LA-101-02	6.02	3.18	2019-10-18	5.26	5.23	0.027	0.006	>50	2030-11-04	
4-LA-101-03	6.02	3.18	2019-10-18	5.38	5.19	0.027	0.038	>50	2030-11-04	
4-LA-101-04	6.02	3.18	2019-10-18	5.29	5.13	0.033	0.032	>50	2030-11-04	
4-LA-101-05	6.02	3.18	2019-10-18	5.38	5.25	0.027	0.026	>50	2030-11-04	
4-LA-101-06	6.02	3.18	2019-10-18	5.22	4.91	0.040	0.061	28.2	2030-11-04	
4-LA-101-07	6.02	3.18	2019-10-18	4.86	4.49	0.052	0.073	17.9	2030-11-04	
4-LA-101-08	6.02	3.18	2019-10-18	5.04	4.70	0.067	0.067	22.6	2030-11-04	
4-LA-101-09	6.02	3.18	2019-10-18	4.86	4.57	0.049	0.057	24.2	2030-11-04	

Line 4-LA-102 | Lean amine supply | A106 Gr.B | t_nom 6.02 mm | t-min 3.18 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-LA-102-01	6.02	3.18	2019-10-18	5.09	5.03	0.035	0.012	>50	2030-11-04	
4-LA-102-02	6.02	3.18	2019-10-18	5.15	5.08	0.035	0.014	>50	2030-11-04	
4-LA-102-03	6.02	3.18	2019-10-18	5.15	4.93	0.040	0.044	40.2	2030-11-04	
4-LA-102-04	6.02	3.18	2019-10-18	4.65	4.40	0.055	0.050	22.0	2030-11-04	
4-LA-102-05	6.02	3.18	2019-10-18	5.23	5.06	0.033	0.034	>50	2030-11-04	
4-LA-102-06	6.02	3.18	2019-10-18	5.52	5.42	0.020	0.020	>50	2030-11-04	
4-LA-102-07	6.02	3.18	2019-10-18	5.32	5.13	0.030	0.038	>50	2030-11-04	
4-LA-102-08	6.02	3.18	2019-10-18	5.55	5.38	0.037	0.034	>50	2030-11-04	

Line 6-LA-103 | Lean amine header | A106 Gr.B | t_nom 7.11 mm | t-min 3.56 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
6-LA-103-01	7.11	3.56	2019-10-18	6.40	6.27	0.026	0.026	>50	2030-11-04	
6-LA-103-02	7.11	3.56	2019-10-18	6.49	6.28	0.032	0.042	>50	2030-11-04	
6-LA-103-03	7.11	3.56	2019-10-18	6.23	5.95	0.039	0.055	43.1	2030-11-04	
6-LA-103-04	7.11	3.56	2019-10-18	5.85	5.64	0.053	0.042	39.0	2030-11-04	
6-LA-103-05	7.11	3.56	2019-10-18	5.81	5.43	0.061	0.075	24.8	2030-11-04	
6-LA-103-06	7.11	3.56	2019-10-18	5.46	5.10	0.072	0.071	21.4	2030-11-04	
6-LA-103-07	7.11	3.56	2019-10-18	5.93	5.73	0.051	0.040	42.9	2030-11-04	
6-LA-103-08	7.11	3.56	2019-10-18	6.19	—	—	—	—	—	inaccessible – no reading
6-LA-103-09	7.11	3.56	2019-10-18	6.16	5.92	0.048	0.048	49.6	2030-11-04	
6-LA-103-10	7.11	3.56	2019-10-18	6.00	5.74	0.064	0.051	34.0	2030-11-04	

Line 4-RA-101 | Rich amine | A106 Gr.B | t_nom 8.56 mm | t-min 3.81 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-RA-101-01	8.56	3.81	2022-09-16	6.83	6.50	0.070	0.156	17.2	2027-11-04	
4-RA-101-02	8.56	3.81	2022-09-16	6.79	6.46	0.073	0.184	14.4	2027-11-04	
4-RA-101-03	8.56	3.81	2022-09-16	5.98	5.52	0.108	0.222	7.7	2027-11-04	gauge photo attached
4-RA-101-04	8.56	3.81	2022-09-16	7.21	6.90	0.060	0.127	24.4	2027-11-04	
4-RA-101-05	8.56	3.81	2022-09-16	6.73	6.37	0.078	0.160	16.0	2027-11-04	
4-RA-101-06	8.56	3.81	2019-10-18	6.94	3.59	0.427	0.664	expired	immediate	< t-min; gauge photo attached
4-RA-101-07	8.56	3.81	2022-09-16	7.50	7.21	0.048	0.093	36.5	2027-11-04	
4-RA-101-08	8.56	3.81	2022-09-16	5.55	5.06	0.126	0.275	4.5	2027-02-10	gauge photo attached
4-RA-101-09	8.56	3.81	2022-09-16	7.37	7.17	0.049	0.101	33.3	2027-11-04	
4-RA-101-10	8.56	3.81	2022-09-16	7.19	6.91	0.061	0.127	24.5	2027-11-04	
4-RA-101-11	8.56	3.81	2022-09-16	6.07	5.53	0.110	0.224	7.7	2027-11-04	gauge photo attached
4-RA-101-12	8.56	3.81	2022-09-16	6.41	6.10	0.184	0.184	12.4	2027-11-04	

Line 4-RA-102 | Rich amine | A106 Gr.B | t_nom 8.56 mm | t-min 3.81 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-RA-102-01	8.56	3.81	2022-09-16	7.42	7.33	0.042	0.050	>50	2027-11-04	
4-RA-102-02	8.56	3.81	2022-09-16	6.43	5.99	0.090	0.192	11.3	2027-11-04	
4-RA-102-03	8.56	3.81	2022-09-16	6.56	6.24	0.081	0.160	15.1	2027-11-04	
4-RA-102-04	8.56	3.81	2022-09-16	6.64	6.38	0.075	0.131	19.7	2027-11-04	
4-RA-102-05	8.56	3.81	2022-09-16	6.56	6.23	0.081	0.164	14.7	2027-11-04	
4-RA-102-06	8.56	3.81	2022-09-16	6.65	6.35	0.075	0.158	16.0	2027-11-04	
4-RA-102-07	8.56	3.81	2022-09-16	6.85	6.53	0.072	0.153	17.8	2027-11-04	
4-RA-102-08	8.56	3.81	2022-09-16	5.86	5.26	0.113	0.226	6.4	2027-11-04	gauge photo attached
4-RA-102-09	8.56	3.81	2022-09-16	6.42	6.04	0.088	0.147	15.2	2027-11-04	
4-RA-102-10	8.56	3.81	2022-09-16	6.86	6.56	0.158	0.158	17.4	2027-11-04	
4-RA-102-11	8.56	3.81	2022-09-16	7.33	7.19	0.073	0.105	32.2	2027-11-04	

Line 6-RA-103 | Rich amine header | A106 Gr.B | t_nom 10.97 mm | t-min 4.57 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
6-RA-103-01	10.97	4.57	2022-09-16	9.55	9.28	0.060	0.109	43.2	2027-11-04	
6-RA-103-02	10.97	4.57	2022-09-16	7.97	7.33	0.131	0.254	10.9	2027-11-04	
6-RA-103-03	10.97	4.57	2022-09-16	8.95	8.44	0.089	0.208	18.6	2027-11-04	
6-RA-103-04	10.97	4.57	2022-09-16	7.30	6.60	0.153	0.321	6.3	2027-11-04	gauge photo attached
6-RA-103-05	10.97	4.57	2022-09-16	7.43	6.91	0.141	0.281	8.3	2027-11-04	gauge photo attached
6-RA-103-06	10.97	4.57	2022-09-16	9.45	9.16	0.065	0.125	36.8	2027-11-04	
6-RA-103-07	10.97	4.57	2022-09-16	9.15	8.89	0.073	0.139	31.2	2027-11-04	

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
6-RA-103-08	10.97	4.57	2022-09-16	9.29	9.07	0.066	0.131	34.4	2027-11-04	
6-RA-103-09	10.97	4.57	2022-09-16	9.43	9.10	0.067	0.135	33.6	2027-11-04	
6-RA-103-10	10.97	4.57	2022-09-16	7.68	7.00	0.140	0.271	9.0	2027-11-04	gauge photo attached
6-RA-103-11	10.97	4.57	2019-10-18	8.91	5.92	0.413	0.592	2.3	2025-12-25	gauge photo attached
6-RA-103-12	10.97	4.57	2022-09-16	9.11	8.74	0.154	0.154	27.0	2027-11-04	
6-RA-103-13	10.97	4.57	2022-09-16	9.13	8.92	0.112	0.125	34.9	2027-11-04	
6-RA-103-14	10.97	4.57	2022-09-16	7.34	6.52	0.345	0.345	5.7	2027-09-02	gauge photo attached

Line 3-RV-101 | Regenerator overhead | A106 Gr.B | t_nom 7.62 mm | t-min 3.56 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
3-RV-101-01	7.62	3.56	2022-09-16	5.92	5.74	0.065	0.061	33.8	2027-11-04	
3-RV-101-02	7.62	3.56	2022-09-16	5.81	5.70	0.065	0.053	32.8	2027-11-04	
3-RV-101-03	7.62	3.56	2022-09-16	4.98	4.72	0.098	0.103	11.3	2027-11-04	
3-RV-101-04	7.62	3.56	2022-09-16	5.39	5.18	0.084	0.079	19.3	2027-11-04	
3-RV-101-05	7.62	3.56	2022-09-16	5.91	5.84	0.064	0.053	35.7	2027-11-04	
3-RV-101-06	7.62	3.56	2022-09-16	5.39	5.22	0.083	0.065	20.1	2027-11-04	
3-RV-101-07	7.62	3.56	2022-09-16	4.80	4.63	0.111	0.087	9.6	2027-11-04	gauge photo attached
3-RV-101-08	7.62	3.56	2022-09-16	5.81	5.64	0.076	0.044	27.5	2027-11-04	

Line 2-RV-102 | Regenerator overhead | A106 Gr.B | t_nom 5.54 mm | t-min 2.79 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
2-RV-102-01	5.54	2.79	2022-09-16	3.91	3.76	0.062	0.079	12.2	2027-11-04	
2-RV-102-02	5.54	2.79	2022-09-16	3.64	3.49	0.073	0.073	9.6	2027-11-04	gauge photo attached
2-RV-102-03	5.54	2.79	2022-09-16	4.34	4.23	0.044	0.028	33.0	2027-11-04	
2-RV-102-04	5.54	2.79	2022-09-16	4.47	4.39	0.042	0.040	38.5	2027-11-04	
2-RV-102-05	5.54	2.79	2022-09-16	4.59	4.57	0.033	0.042	42.8	2027-11-04	
2-RV-102-06	5.54	2.79	2022-09-16	4.05	3.94	0.062	0.065	17.6	2027-11-04	

Line 3-RF-101 | Reflux | A312 TP316 | t_nom 5.49 mm | t-min 2.41 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
3-RF-101-01	5.49	2.41	2019-10-18	5.40	5.15	0.012	0.050	>50	2030-11-04	
3-RF-101-02	5.49	2.41	2019-10-18	5.38	5.24	0.007	0.028	>50	2030-11-04	
3-RF-101-03	5.49	2.41	2019-10-18	5.35	5.35	0.003	0.000	>50	2030-11-04	
3-RF-101-04	5.49	2.41	2019-10-18	5.29	5.21	0.009	0.016	>50	2030-11-04	
3-RF-101-05	5.49	2.41	2019-10-18	5.17	5.14	0.011	0.006	>50	2030-11-04	
3-RF-101-06	5.49	2.41	2019-10-18	5.12	5.13	0.014	-0.002	>50	2030-11-04	
3-RF-101-07	5.49	2.41	2019-10-18	5.38	5.25	0.006	0.026	>50	2030-11-04	
3-RF-101-08	5.49	2.41	2019-10-18	5.23	5.12	0.022	0.022	>50	2030-11-04	
3-RF-101-09	5.49	2.41	2019-10-18	5.13	5.10	0.002	0.006	>50	2030-11-04	

Line 2-RF-102 | Reflux | A312 TP316 | t_nom 3.91 mm | t-min 2.16 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
2-RF-102-01	3.91	2.16	2019-10-18	3.66	3.55	0.013	0.022	>50	2030-11-04	
2-RF-102-02	3.91	2.16	2019-10-18	3.65	3.53	0.012	0.024	>50	2030-11-04	
2-RF-102-03	3.91	2.16	2019-10-18	3.68	3.65	0.010	0.006	>50	2030-11-04	
2-RF-102-04	3.91	2.16	2019-10-18	2.83	1.94	0.067	0.176	expired	immediate	< t-min; gauge photo attached
2-RF-102-05	3.91	2.16	2019-10-18	2.88	1.92	0.159	0.190	expired	immediate	< t-min; gauge photo attached
2-RF-102-06	3.91	2.16	2019-10-18	2.74	1.92	0.160	0.162	expired	immediate	< t-min; gauge photo attached

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
2-RF-102-07	3.91	2.16	2019-10-18	2.82	1.86	0.073	0.190	expired	immediate	< t-min; gauge photo attached

Line 4-LA-104 | L/R exchanger, lean side | A106 Gr.B | t_nom 6.02 mm | t-min 3.18 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-LA-104-01	6.02	3.18	2019-10-18	5.16	5.02	0.037	0.028	>50	2030-11-04	
4-LA-104-02	6.02	3.18	2019-10-18	4.79	4.55	0.052	0.048	26.5	2030-11-04	
4-LA-104-03	6.02	3.18	2019-10-18	5.23	5.20	0.027	0.006	>50	2030-11-04	
4-LA-104-04	6.02	3.18	2019-10-18	5.49	5.34	0.025	0.030	>50	2030-11-04	
4-LA-104-05	6.02	3.18	2019-10-18	4.71	4.37	0.058	0.067	17.7	2030-11-04	
4-LA-104-06	6.02	3.18	2019-10-18	5.38	5.17	0.032	0.042	47.8	2030-11-04	
4-LA-104-07	6.02	3.18	2019-10-18	5.15	4.92	0.038	0.046	38.2	2030-11-04	
4-LA-104-08	6.02	3.18	2019-10-18	4.88	4.63	0.046	0.050	29.3	2030-11-04	
4-LA-104-09	6.02	3.18	2019-10-18	4.61	4.37	0.048	0.048	25.0	2030-11-04	
4-LA-104-10	6.02	3.18	2019-10-18	4.63	4.32	0.060	0.061	18.6	2030-11-04	

Line 4-RA-104 | L/R exchanger, rich side | A106 Gr.B | t_nom 8.56 mm | t-min 3.81 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-RA-104-01	8.56	3.81	2022-09-16	6.61	6.33	0.076	0.162	15.5	2027-11-04	
4-RA-104-02	8.56	3.81	2022-09-16	7.04	6.84	0.060	0.093	32.5	2027-11-04	
4-RA-104-03	8.56	3.81	2022-09-16	7.70	7.67	0.065	0.053	>50	2027-11-04	
4-RA-104-04	8.56	3.81	2022-09-16	7.31	6.98	0.109	0.131	24.2	2027-11-04	
4-RA-104-05	8.56	3.81	2022-09-16	7.12	6.74	0.119	0.164	17.8	2027-11-04	
4-RA-104-06	8.56	3.81	2022-09-16	6.70	6.41	0.076	0.143	18.2	2027-11-04	
4-RA-104-07	8.56	3.81	2022-09-16	6.04	5.61	0.104	0.210	8.6	2027-11-04	gauge photo attached
4-RA-104-08	8.56	3.81	2022-09-16	7.09	6.90	0.093	0.111	27.9	2027-11-04	

Line 2-ST-101 | Reboiler steam | A106 Gr.B | t_nom 3.91 mm | t-min 2.29 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
2-ST-101-01	3.91	2.29	2019-10-18	3.59	3.42	0.018	0.034	33.6	2030-11-04	
2-ST-101-02	3.91	2.29	2019-10-18	3.34	3.19	0.025	0.030	30.3	2030-11-04	
2-ST-101-03	3.91	2.29	2019-10-18	3.68	—	—	—	—	—	inaccessible – no reading
2-ST-101-04	3.91	2.29	2019-10-18	3.52	3.50	0.011	0.004	>50	2030-11-04	
2-ST-101-05	3.91	2.29	2019-10-18	3.27	3.22	0.024	0.010	38.6	2030-11-04	
2-ST-101-06	3.91	2.29	2019-10-18	3.66	3.66	-0.003	0.000	>50	2030-11-04	

Line 3-CW-101 | Amine cooler CW | A106 Gr.B | t_nom 5.49 mm | t-min 2.79 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
3-CW-101-01	5.49	2.79	2019-10-18	4.91	4.79	0.022	0.024	>50	2030-11-04	
3-CW-101-02	5.49	2.79	2019-10-18	4.98	4.69	0.025	0.057	33.1	2030-11-04	
3-CW-101-03	5.49	2.79	2019-10-18	4.87	4.63	0.031	0.048	38.7	2030-11-04	
3-CW-101-04	5.49	2.79	2019-10-18	5.20	5.22	0.008	-0.004	>50	2030-11-04	
3-CW-101-05	5.49	2.79	2019-10-18	5.14	5.19	0.010	-0.010	>50	2030-11-04	
3-CW-101-06	5.49	2.79	2019-10-18	5.01	4.91	0.019	0.020	>50	2030-11-04	

Line 3-CW-102 | Amine cooler CW | A106 Gr.B | t_nom 5.49 mm | t-min 2.79 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
3-CW-102-01	5.49	2.79	2019-10-18	4.85	4.77	0.025	0.016	>50	2030-11-04	
3-CW-102-02	5.49	2.79	2019-10-18	4.17	4.09	0.050	0.016	26.2	2030-11-04	
3-CW-102-03	5.49	2.79	2019-10-18	4.27	4.12	0.049	0.030	27.0	2030-11-04	

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
3-CW-102-04	5.49	2.79	2019-10-18	4.50	4.29	0.043	0.042	35.2	2030-11-04	
3-CW-102-05	5.49	2.79	2019-10-18	4.45	4.35	0.036	0.020	43.9	2030-11-04	

Sign-off	Name	Function	Date	Signature
Prepared	Marcus Chen	lead UT technician, SNT-TC-1A UT/MT/PT/VT Level II; CAN/CGSB-48.9712 UT Level II	2024-11-21	_____
Reviewed	David Park	Level III / Quality Manager, SNT-TC-1A UT Level III	2024-11-21	_____